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Exp No:-1

Study of various Network commands used in Linux and Windows

AIM: Study of various Network commands used in Linux and Windows

Basic Networking commands:

ARP.

arp -a: ARP is a short form of adress resaultion protocol It will show the IP adress of yours computer along with the IP address and MAC adress of your router.

```
C:\Windows\system32>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.

C:\Windows\system32>
```

2) Hostname

Displays & changes the name of the local computer

```
C:\Windows\system32>hostname
DESKTOP-G0LVE4J

C:\Windows\system32>
```

3) IPconfig (windows) / ifconfig (Linux)

Display and configures the IP adress sub network mask and default gateway.

```
C:\Windows\system32>ipconfig

Windows IP Configuration


Ethernet adapter Npcap Loopback Adapter:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::b923:c6f7:7122:8700%6
   Autoconfiguration IPv4 Address. . : 169.254.158.128
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::390c:dc1c:3ca7:94bc%3
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   IPv4 Address. . . . . . . . . . . : 172.16.10.86
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : 172.16.8.1

C:\Windows\system32>
```

4. NBTstat (Net Bios Statistics)

Displays NetBIOS over TCP/IP protocol Statisti and NetBIOS name tables

```
C:\Windows\system32>nbtstat

Displays protocol statistics and current TCP/IP connections using NBT
(NetBIOS over TCP/IP).

NBTSTAT [ [-a RemoteName] [-A IP address] [-c] [-n]
        [-r] [-R] [-RR] [-s] [-S] [interval] ]

   -a   (adapter status) Lists the remote machine's name table given its name
   -A   (Adapter status) Lists the remote machine's name table given its
                         IP address.
   -c   (cache)          Lists NBT's cache of remote [machine] names and their IP addresses
   -n   (names)          Lists local NetBIOS names.
   -r   (resolved)       Lists names resolved by broadcast and via WINS
   -R   (Reload)         Purges and reloads the remote cache name table
   -S   (Sessions)       Lists sessions table with the destination IP addresses
   -s   (sessions)       Lists sessions table converting destination IP
                         addresses to computer NETBIOS names.
   -RR  (ReleaseRefresh) Sends Name Release packets to WINS and then, starts Refresh

   RemoteName   Remote host machine name.
   IP address   Dotted decimal representation of the IP address.
   interval     Redisplays selected statistics, pausing interval seconds
                between each display. Press Ctrl+C to stop redisplaying
                statistics.


C:\Windows\system32>
```

5) Net stat

Displays active network. connections routing tables and listening

```
C:\Windows\system32>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49675        DESKTOP-G0LVE4J:49676  ESTABLISHED
  TCP    127.0.0.1:49676        DESKTOP-G0LVE4J:49675  ESTABLISHED
  TCP    127.0.0.1:49677        DESKTOP-G0LVE4J:49678  ESTABLISHED
  TCP    127.0.0.1:49678        DESKTOP-G0LVE4J:49677  ESTABLISHED
  TCP    172.16.10.86:13133     4.213.25.242:https     ESTABLISHED
  TCP    172.16.10.86:13199     lcbomp-in-f188:5228    ESTABLISHED
  TCP    172.16.10.86:13258     a23-202-229-23:https   CLOSE_WAIT
  TCP    172.16.10.86:13259     150.171.110.82:https   CLOSE_WAIT
  TCP    172.16.10.86:13260     131.253.33.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13261     135.233.41.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13262     204.79.197.222:https   CLOSE_WAIT
  TCP    172.16.10.86:13270     72.154.187.226:https   CLOSE_WAIT
  TCP    172.16.10.86:13272     20.100.121.19:https    CLOSE_WAIT
  TCP    172.16.10.86:13273     a-0003:http            CLOSE_WAIT
  TCP    172.16.10.86:13278     13.107.42.254:https    CLOSE_WAIT
  TCP    172.16.10.86:13279     52.108.9.254:https     CLOSE_WAIT
  TCP    172.16.10.86:13281     172.202.65.254:https   CLOSE_WAIT
  TCP    172.16.10.86:13293     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13294     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13295     a23-211-142-112:https  TIME_WAIT
  TCP    172.16.10.86:13297     bkk02s04-in-f14:https  ESTABLISHED
  TCP    172.16.10.86:13298     4.213.25.242:https     ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:1521  DESKTOP-G0LVE4J:49679  ESTABLISHED
  TCP    [fe80::390c:dc1c:3ca7:94bc%3]:49679  DESKTOP-G0LVE4J:1521  ESTABLISHED

C:\Windows\system32>
```

6) NS Lookup

Queries DNS servers to find IP address of domain of a host

```
C:\Windows\system32>nslookup www.google.com
Server:  UnKnown
Address:  172.16.8.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:4827:7700::
          2001:4860:482b:7700::
          2001:4860:4828:7700::
          2001:4860:482a:7700::
          2001:4860:4826:7700::
          2001:4860:4829:7700::
          2001:4860:482d:7700::
          2001:4860:482c:7700::
          142.251.156.119
          142.251.155.119
          142.251.151.119
          142.251.150.119
          142.251.157.119
          142.251.152.119
          142.251.154.119
          142.251.153.119
```

7. Path ping

combines the features of ping and Traceroutes to identify packet loss and network latency

```
C:\Windows\system32>pathping

Usage: pathping [-g host-list] [-h maximum_hops] [-i address] [-n]
                [-p period] [-q num_queries] [-w timeout]
                [-4] [-6] target_name

Options:
    -g host-list        Loose source route along host-list.
    -h maximum_hops     Maximum number of hops to search for target.
    -i address          Use the specified source address.
    -n                  Do not resolve addresses to hostnames.
    -p period           Wait period milliseconds between pings.
    -q num_queries      Number of queries per hop.
    -w timeout          Wait timeout milliseconds for each reply.
    -4                  Force using IPv4.
    -6                  Force using IPv6.

C:\Windows\system32>
```

8. ping (packet Internet Groper)

Tasks network connectivity between two devices by ICMP echo requests

```
C:\Windows\system32>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

9) Route

Displays or modifies the IP routing table used for packet forwarding

```
C:\Windows\system32>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f            Clears the routing tables of all gateway entries.  If this is
                used in conjunction with one of the commands, the tables are
                cleared prior to running the command.

  -p            When used with the ADD command, makes a route persistent across
                boots of the system. By default, routes are not preserved
                when the system is restarted. Ignored for all other commands,
                which always affect the appropriate persistent routes.

  -4            Force using IPv4.

  -6            Force using IPv6.

  command       One of these:
                  PRINT     Prints  a route
                  ADD       Adds    a route
                  DELETE    Deletes a route
                  CHANGE    Modifies an existing route
  destination   Specifies the host.
  MASK          Specifies that the next parameter is the 'netmask' value.
  netmask       Specifies a subnet mask value for this route entry.
                If not specified, it defaults to 255.255.255.255.
  gateway       Specifies gateway.
  interface     the interface number for the specified route.
  METRIC        specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway are looked up in the host name
database file HOSTS.

If the command is PRINT or DELETE. Destination or gateway can be a wildcard,
(wildcard is specified as a star '*'), or the gateway argument may be omitted.

If Dest contains a * or ?, it is treated as a shell pattern, and only
matching destination routes are printed. The '*' matches any string,
and '?' matches any one char. Examples: 157.*.1, 157.*, 127.*, *224*.

Pattern match is only allowed in PRINT command.
Diagnostic Notes:
    Invalid MASK generates an error, that is when (DEST & MASK) != DEST.
    Example> route ADD 157.0.0.0 MASK 155.0.0.0 157.55.80.1 IF 1
             The route addition failed: The specified mask parameter is invalid. (Destination & Mask) != Destination.

Examples:

    > route PRINT
    > route PRINT -4
    > route PRINT -6
    > route PRINT 157*                .... Only prints those matching 157*

    > route ADD 157.0.0.0 MASK 255.0.0.0  157.55.80.1 METRIC 3 IF 2
            destination^      ^mask      ^gateway     metric^    ^
                                                      Interface^
      If IF is not given, it tries to find the best interface for a given
      gateway.
    > route ADD 3ffe::/32 3ffe::1

    > route CHANGE 157.0.0.0 MASK 255.0.0.0 157.55.80.5 METRIC 2 IF 2

      CHANGE is used to modify gateway and/or metric only.

    > route DELETE 157.0.0.0
    > route DELETE 3ffe::/32

C:\Windows\system32>_
```

Result :-

The networking commands were successfully executed and studied their outputs were observed and used to understand network config connectivity testing DNS resolution and trouble shooting.